LLM Trust Score: Scoring the Reliability of Language-Model Answers with Cross-Model Judges and Self-Consistency

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Abstract

Large language models (LLMs) often give fluent answers that are wrong, and users have no cheap way to tell which answers to trust. This paper presents LLM Trust Score, a black-box layer that scores one answer from 0 to 100 without access to token probabilities. A writer model (gpt-4o-mini) answers each question three times at temperature 0.9. A Gemini model checks whether the three answers reach the same conclusion. Three judges from three other developers (Qwen, Mistral and Llama) rate each answer as correct or hallucinated with the help of web-search snippets, and the best-scoring answer receives a score and a High, Medium or Low label. A separate tie-breaker model (NVIDIA Nemotron) checks only the winning answer and removes 10 points when it disagrees with the judge majority. We evaluate on 200 fresh questions (66 TruthfulQA, 66 HaluEval and 68 TriviaQA) graded by Claude Haiku 5.5, a model that plays no part in the scoring. The writer was correct on 61.0% of questions. The trust score separated correct from incorrect answers with an AUROC of 0.674 (95% bootstrap interval 0.604–0.740) and a mean gap of 18.1 points (11.1–25.5). The labels were ordered as intended: 71.1% of High, 55.0% of Medium and 26.3% of Low answers were correct. The score is not a calibrated probability. Of the answers that scored 100, 28.0% were wrong, and the Brier score (0.272) was worse than always predicting the base rate (0.238). Discrimination was weakest on TruthfulQA (AUROC 0.568, interval 0.450–0.682). A simple proxy that uses only the judges' hallucination flags ranked answers at least as well as the full score (AUROC 0.721). Reading the tie-breaker's penalty back from the scores, it disagreed with the judges on 17 answers (8.5%), of which 35.3% were correct against 63.4% for the rest, and it added only 0.013 to the AUROC (interval −0.003 to +0.033). We give the reasons for every design choice, including why a Gemini agreement checker was preferred to a natural-language-inference (NLI) model, and we state which components were not evaluated on their own.

Keywords: large language models; hallucination detection; LLM-as-a-judge; self-consistency; confidence estimation; retrieval grounding; TruthfulQA

1. Introduction

1.1 Background and Motivation

Large language models (LLMs) now answer questions on medicine, law, science and everyday facts, and they do so in a confident, fluent style whether or not the answer is correct. The research literature calls an answer that is fluent but unsupported or false a hallucination, and surveys describe it as one of the main obstacles to using these models where mistakes matter (Ji et al., 2023; Huang et al., 2025). The problem has a theoretical side as well. Kalai and Vempala (2024) show that a language model that is well calibrated as a predictor of text must still hallucinate at roughly the rate at which the facts it was trained on appear only once, even when the training data contain no errors. They also stress that hallucination is not inevitable for facts that are seen repeatedly, and that post-training may reduce it. In practice, hallucination remains a property of deployed systems, not a solved problem.

Farquhar et al. (2024) describe one important kind of hallucination, which they call confabulation: an answer that is arbitrary and incorrect, and that changes from one sampling of the model to the next. They note that confident false outputs block adoption in areas such as law and medicine, and they show that the amount of variation across sampled answers, measured over meaning and not over wording, can flag such errors. The authors are explicit that this addresses only one error type. A model can also be consistently wrong, because its training data contain a false belief, and sampling alone cannot expose that.

A user who receives a single answer from a chat system has no way of knowing which of these situations applies. What the user needs is a per-answer reliability signal that is cheap to compute, that works with models offered only through an API, and that can be explained in plain language. Many hosted models do not expose token probabilities, so a useful signal has to be computed from outputs alone. Xiong et al. (2024) evaluated such black-box confidence methods on five dataset types and found that they reach failure-prediction performance close to white-box methods (AUROC 0.522 against 0.605 in their comparison), although none of the methods was consistently best and all of them struggled on tasks that need professional knowledge. This paper builds on that finding.

1.2 Problem Statement

Given a factual question, the aim is to return one answer and a trust score between 0 and 100 that is higher when the answer is more likely to be correct. The setting has four constraints. First, the layer must be black-box: it may call models but may not read their internal state. Second, it must not need a labelled answer at the time of use. Third, its cost and delay must be small enough for an interactive web page. Fourth, its output must be explainable, so that a user can see how many models confirmed the answer and whether the answers agreed with each other.

Existing approaches each use one source of evidence. Self-consistency methods compare several samples from one model (Wang et al., 2023; Manakul et al., 2023; Kuhn et al., 2023). LLM-as-a-judge methods ask a second model for a verdict (Zheng et al., 2023). Each source

has a known weakness. Samples from one model share that model's beliefs, and a single judge has its own biases, among them a preference for its own writing (Panickssery et al., 2024). Combining sources that fail in different ways is the motivation for the layer studied here.

1.3 Aim and Objectives

The aim is to design, implement and critically evaluate a black-box trust layer that combines agreement between sampled answers with the verdicts of independent judges, and to state honestly what the resulting score can and cannot be used for. The objectives are:

- to specify a pipeline in which the model that writes the answers, the model that checks agreement, the judges, the tie-breaker and the grader are all different models, so that no model checks its own work;
- to give, for every design decision, the published evidence that motivates it and to say plainly where no direct evidence exists;
- to evaluate the layer on 200 questions that were not used while developing it, with confidence intervals;
- to test whether the score ranks correct answers above incorrect ones, whether the High, Medium and Low labels are ordered, and whether the score behaves like a probability; and
- to document the limitations of the design and of the evaluation, including the components that were not evaluated separately.

1.4 Research Questions

- RQ1: Does the trust score separate correct answers from incorrect answers, and by how much?
- RQ2: Does the separation hold across benchmarks with different kinds of error (TriviaQA, HaluEval and TruthfulQA)?
- RQ3: Are the confidence labels useful, and can the score be read as a probability of being correct?
- RQ4: Which part of the pipeline carries the signal, and what can be said about the parts that were not isolated?

1.5 Contributions

The paper makes four contributions.

- A fully specified black-box pipeline in which the writer, agreement checker, judges, tie-breaker and grader all come from different developers, together with the evidence for and against each choice (Section 3 and Table 3.1).

- A reasoned comparison of an LLM agreement checker with an NLI checker, supported by the results of Manakul et al. (2023), and an explicit statement that the two were not compared on our own data.
- An evaluation on 200 fresh questions with bootstrap intervals, in which the main finding is moderate discrimination (AUROC 0.674) but poor calibration, with no claim beyond that.
- An open implementation and a public web page that serve the same pipeline (Section 3.10).

1.6 Structure of the Paper

Section 2 reviews related work. Section 3 describes the method and the evaluation protocol. Section 4 reports the results. Section 5 discusses them, Section 6 states the limitations, and Section 7 concludes with future work.

2. Related Work

2.1 Hallucination in Language Models

Ji et al. (2023) survey hallucination in natural language generation, define it as generated content that is nonsensical or unfaithful to the source, and organise the field by task and by cause. Huang et al. (2025) give a taxonomy specific to LLMs and review detection and mitigation methods. Benchmarks measure the problem in different ways. TruthfulQA (Lin et al., 2022) contains 817 questions in 38 categories that were written so that some humans would answer falsely because of a misconception. The best model in the original study was truthful on 58% of questions against 94% for humans, and the largest models were generally the least truthful, which the authors explain by models learning false statements that are common in their training text. HaluEval (Li et al., 2023) provides 35,000 samples that were built with a sampling-then-filtering procedure using ChatGPT and human labellers, and it reports that LLMs find it hard to recognise hallucinations unless they are given external knowledge or reasoning steps. TriviaQA (Joshi et al., 2017) is a large question-answering dataset of trivia questions.

2.2 Detection from Sampled Answers

Wang et al. (2023) introduced self-consistency for reasoning: instead of one greedy answer, sample several reasoning paths and keep the most frequent final answer. The method improved accuracy on GSM8K (+17.9 points), AQuA (+12.2), StrategyQA (+6.4) and ARC-challenge (+3.9) across four language models, with no extra training. Its use here is the sampling idea, not the claim that agreement implies truth, which the paper does not make.

SelfCheckGPT (Manakul et al., 2023) turns the sampling idea into a detector for hallucination. The premise is that facts a model knows tend to recur across samples while

invented facts diverge. It needs no external database and no access to model internals, and it was evaluated on passages about people generated by GPT-3 and annotated for factuality. The authors compare five ways of scoring consistency (BERTScore, question answering, n-gram, an NLI model and a prompt to an LLM) and we return to that comparison in Section 3.3.

Kuhn et al. (2023) argue that uncertainty over generated language should be measured over meanings and not over strings, because different sentences can mean the same thing. They group sampled answers into meaning classes using bidirectional entailment, computed with a DeBERTa-large model fine-tuned on MNLI, and compute the entropy over the classes. This semantic entropy predicted accuracy better than the baselines on question-answering data. Farquhar et al. (2024) extended the idea in *Nature* and showed that entropy over meanings flags confabulations across datasets and tasks without task-specific data. Both papers motivate the agreement check in Section 3.3, and Kuhn et al. also record its limitations: equivalence is soft and need not be transitive, so clusters can depend on the order in which pairs are compared.

2.3 Elicited Confidence and Calibration

A second line of work asks the model for its confidence. Tian et al. (2023) studied models tuned with human feedback on TriviaQA, SciQ and TruthfulQA and found that confidences the model states in words are typically better calibrated than its token probabilities, often reducing expected calibration error by a relative 50%. Xiong et al. (2024) found that verbalised confidence is often too high, and that sampling several answers and aggregating their consistency reduces the overconfidence. These results explain why the layer relies on judgments and agreement and not on a stated confidence from the writer.

Calibration itself has a standard vocabulary. Guo et al. (2017) showed that modern neural networks are often miscalibrated and popularised the expected calibration error. The Brier (1950) score measures the squared error of probability forecasts, and Fawcett (2006) describes ROC analysis and the area under the curve (AUROC), which measures how well a score ranks positive cases above negative ones without fixing a threshold. We use these tools in Section 3.9.

2.4 LLM Judges and Their Biases

Zheng et al. (2023) showed that a strong LLM judge reaches more than 80% agreement with human preferences on chat responses, the same level as humans reach with each other. They also documented position bias, verbosity bias and self-enhancement bias (a preference for text the judge wrote), and noted that some of these biases are small or can be reduced. Wang et al. (2024) showed that the ranking of two candidate answers can be changed by swapping their order and proposed calibration methods. Because each judge in our pipeline sees one answer at a time, order effects between candidates do not arise, but the other biases remain possible.

Panickssery et al. (2024) found that GPT-3.5, GPT-4 and Llama 2 favour their own summaries, that GPT-4 recognises its own text 73.5% of the time, and that stronger

self-recognition goes with stronger self-preference. Huang et al. (2024) showed that, without external feedback, LLMs struggle to correct their own reasoning and sometimes become worse. Both findings concern other tasks, summarisation and reasoning, so they motivate keeping the writer, the judges and the grader apart without proving that the same effect occurs for factual verdicts. Kamalloo et al. (2023) examined how to grade open-domain question answering in the era of LLMs. Matching against a list of gold answers underestimates generative models, with one model gaining nearly 60% under human judgment, and automated evaluators help but cannot fully replace human evaluation. This is the basis for using an LLM as the grader in Section 3.9 and the reason that its agreement with people is listed as a limitation.

2.5 Correlated Errors and Multi-Model Checking

Using several models is only useful if their errors differ. Kim et al. (2025) studied more than 350 LLMs and found that, on one leaderboard, two models that are both wrong give the same wrong answer 60% of the time. Shared architecture and shared provider raise the correlation, and larger, more accurate models are more correlated even across providers. The paper also shows the effect on LLM-as-a-judge evaluation. The finding supports choosing judges from different developers, and it also warns that doing so does not remove shared mistakes.

Du et al. (2024) had several instances of a model propose answers and debate them over several rounds, and reported better mathematical and strategic reasoning and fewer factual errors and hallucinations. Their instances mostly came from one model family, so the result motivates cross-checking by several models but is not evidence for cross-lab panels in particular.

2.6 Retrieval Grounding

Retrieval gives a model access to knowledge it does not hold in its parameters. Lewis et al. (2020) combined parametric and non-parametric memory for knowledge-intensive tasks. Shuster et al. (2021) reported that retrieval-augmented dialogue models reduced hallucinated responses by more than 60%, using Wikipedia as the source. Li et al. (2023) found that supplying external knowledge helps LLMs recognise hallucinations. These studies concern generation and dialogue more than verification of short factual answers, so they motivate web grounding of the judges in Section 3.4 without measuring its effect here.

2.7 Research Gap and Positioning

Table 2.1 places the layer among the families above. Sampling methods use one model and measure agreement, judge methods use one model and measure a verdict, and confidence methods rely on what a model says about itself. The layer studied here combines agreement from the writer with verdicts from three other developers, adds web grounding and a fifth independent check, and reports a bounded score with labels. What is missing from the literature, and what this paper tries to supply, is a transparent evaluation of such a combination on fresh questions, with intervals and with a clear statement of which parts were not tested.

Approach	Evidence used	What it does not cover
Self-consistency (Wang et al., 2023)	Agreement among samples from one model	Truth of the agreed answer
SelfCheckGPT (Manakul et al., 2023)	Consistency of sampled passages	Consistently wrong answers
Semantic entropy (Kuhn et al., 2023; Farquhar et al., 2024)	Entropy over meaning classes of samples	False beliefs shared by all samples
Verbalised confidence (Tian et al., 2023; Xiong et al., 2024)	The model's stated confidence	Overconfidence of the stated value
LLM judge (Zheng et al., 2023)	One model's verdict	That model's own biases
This layer	Agreement, three cross-lab judges, web snippets, one extra check	Shared errors across developers (Kim et al., 2025)

Table 2.1 — How the trust layer relates to published approaches.

3. Method

3.1 Overview

For a question q, the pipeline produces one answer, a trust score between 0 and 100, a label and a written reason. It has six steps (Figure 3.1). A writer model produces three independent answers. A second model decides which of those answers reach the same conclusion. For each answer, a web search supplies context and three judges from three other developers return one verdict each. Each answer is scored from the verdicts and the agreement, and the highest-scoring answer is chosen. A further model then checks the chosen answer, and the score is reduced if it disagrees with the judge majority. Finally the score is mapped to a label.

Every role is filled by a model from a different developer (Table 3.1). This separation is the central design rule. It is motivated by the self-preference results of Panickssery et al. (2024) and the self-enhancement bias reported by Zheng et al. (2023), and by the finding of Kim et al. (2025) that errors are more strongly correlated within a provider than across providers. The rule reduces the chance that a model grades its own work. It does not make the models independent, as Section 6 explains.

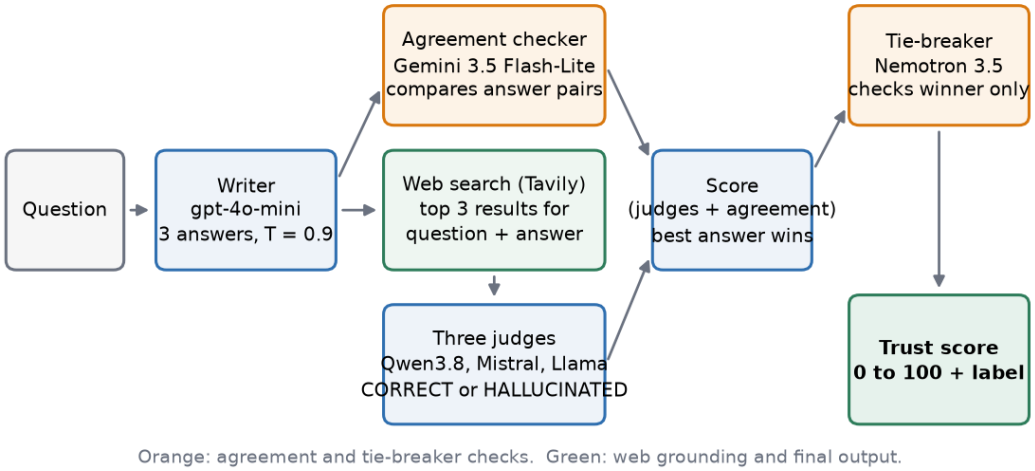


Figure 3.1 — The trust-score pipeline. Orange boxes are the agreement check and the tie-breaker; green boxes are web grounding and the output.

Role	Model (provider)	Developer	Setting
Writer	gpt-4o-mini (OpenAI API)	OpenAI	3 calls, T = 0.9
Agreement checker	Gemini 3.5 Flash-Lite (Gemini API)	Google	T = 0, YES or NO
Judge 1	Qwen3.8-27B (Groq)	Alibaba	T = 0, low reasoning effort
Judge 2	Mistral Medium 3.5 (OpenRouter)	Mistral AI	T = 0, max 512 tokens
Judge 3	Llama 3.3 70B Instruct (OpenRouter)	Meta	T = 0, max 512 tokens
Tie-breaker	Nemotron 3.5 Lightning 30B (NVIDIA API)	NVIDIA	T = 0, winner only
Web search	Tavily, basic depth	Tavily	top 3 results plus summary
Grader (evaluation only)	Claude Haiku 5.5 (OpenRouter)	Anthropic	T = 0, YES or NO

Table 3.1 — Models, roles and settings used in the evaluated configuration.

3.2 Answer Generation

The writer, gpt-4o-mini, receives the question and answers it three times at temperature 0.9, each call independent of the others. Sampling at a temperature above zero follows the practice of Wang et al. (2023), who sample diverse outputs and compare them, and of Manakul et al. (2023), who use sampled passages as evidence about the main answer. A system instruction, sent with every model call in the pipeline, tells the model that when a question concerns something that does not exist, such as a fake study, person or event, it must say so and stop, without inventing details. If a generation call fails, the pipeline continues with the answers that were returned, and if all three fail it returns a score of zero with an error label.

Three samples is a small number. Manakul et al. (2023) report results with larger numbers of samples, and the value of three here is a cost decision made because every extra sample multiplies the number of judge calls (Section 3.10). Section 6 lists this as a limitation.

3.3 Agreement Check

3.3.1 What the check does

The agreement check supplies the self-consistency signal. The checker receives two answers and is asked whether they reach the *same final conclusion*. The prompt states that extra explanation or detail in one answer does not count as disagreement, and it gives a worked example in which a bare number and a step-by-step derivation of the same number agree. The model must reply with one word, YES or NO, at temperature 0. Answers are then grouped greedily. Answer 1 is compared with answers 2 and 3, the first answer not yet assigned starts a new group, and so on, which costs two or three checker calls per question. For one answer, the consistency fraction is the share of the *other* answers that fall in its group. With three answers it takes the values 0, 0.5 or 1.

The design follows the idea of Kuhn et al. (2023) and Farquhar et al. (2024), who group sampled answers into meaning classes using bidirectional entailment and measure the spread of those classes. We use the group sizes in a simpler way, as the share of agreeing answers, because the score must be easy to explain. An entropy value over the groups is also computed and shown to the user, but it is not part of the trust score.

3.3.2 Why a Gemini checker and not an NLI model

The obvious alternative is the NLI approach of Kuhn et al. (2023), who check entailment in both directions with a DeBERTa-large model fine-tuned on MNLI, with the question added as context because a short reply such as “Paris.” does not entail “The capital of France is Paris” without it. We chose an LLM prompt instead, for four reasons, the first of which is the strongest.

- **Published evidence favours the prompted LLM for consistency scoring.** Manakul et al. (2023) compared an NLI variant of SelfCheckGPT with a variant that prompts an LLM (ChatGPT) to judge consistency. In their Table 2 the prompted variant scored higher on every reported metric: sentence-level AUC-PR of 93.42 against 92.50 on non-factual sentences, 53.19 against 45.17 on the harder non-factual subset and 67.09 against 66.08 on factual sentences, and a passage-level Pearson correlation of 78.32 against 74.14. The data were GPT-3 passages about people, so the comparison is indirect for short factual answers, but it is the closest published test of the choice that we know of.
- **Our answers are short statements with variable extra detail.** A reply such as “Bell, who received the first US patent for the telephone in 1876” and a reply that is only “Bell” reach the same conclusion. A strict two-way entailment test can label such a pair neutral and so count it as disagreement, which would lower the consistency score for the wrong reason. The prompt states the rule that extra detail is not disagreement. This reasoning is a hypothesis. It was not tested on our data (Section 6).
- **Deployment weight.** An NLI classifier needs a deep-learning runtime and a model file of about one gigabyte on the server, and it slows the first request of the public web page. The Gemini checker is an API call.

- **Role separation.** Gemini is from a developer that plays no other role in the pipeline, which keeps the checker from agreeing with answers that resemble its own output (Panickssery et al., 2024).

The choice has costs that we accept and state. An LLM checker is not perfectly repeatable and adds a small API cost per question, while an NLI model is free to run and deterministic. Kuhn et al. (2023) validated their NLI checker manually and reported high agreement with people on short TriviaQA and CoQA answers (about 93% on TriviaQA, taken from their appendix). We have not run an equivalent validation for the Gemini checker, so the superiority of the LLM checker for our data is a design judgment supported by indirect evidence and is not a measured result. Hand labelling 30 answer sets and comparing both checkers would turn it into one.

3.4 Web Grounding and the Judges

For each answer the pipeline runs one web search with the query “question plus answer”, using Tavily at basic depth, and keeps the generated summary and the titles and contents of the top three results. This context is added to the prompt of every judge for that answer. If the search fails, the judges receive the question and answer alone. The motivation is the evidence that retrieval reduces hallucination in generation (Shuster et al., 2021; Lewis et al., 2020) and that external knowledge helps LLMs recognise hallucinations (Li et al., 2023). Section 6 notes that web grounding was not switched off in a separate run, so its effect on the score is not measured.

Each judge sees the question, the proposed answer, the search context and one instruction: based on its own knowledge and the context, is the answer factually correct and not hallucinated? It must reply with one word, CORRECT or HALLUCINATED, at temperature 0. The reply is read with a conservative parser. Negations such as HALLUCINATED, INCORRECT, NOT CORRECT, NOT ACCURATE and WRONG are tested first and return a negative verdict, only then is the word CORRECT accepted, and an unreadable reply makes the judge count as failed. This order avoids a known pitfall, because a plain substring test for CORRECT would also accept INCORRECT.

The panel has three judges from three developers: Qwen3.8-27B (Alibaba, through Groq), Mistral Medium 3.5 (Mistral AI) and Llama 3.3 70B (Meta). Three developers are used because Kim et al. (2025) found that error correlation rises with shared provider and architecture, so judges from one provider would add less information. The same paper found that accurate models from different providers are still correlated, which is why the panel cannot be described as independent. Judging one answer at a time, not comparing two, means that the position bias reported by Wang et al. (2024) does not apply, while the other biases listed by Zheng et al. (2023) remain possible. When a judge fails, for example because of a rate limit, the pipeline retries with waiting and drops the judge if it still fails, so that one failed call does not stop a question.

3.5 Scoring

Let m be the number of judges that responded for answer i, let v_{ij} be 1 when judge j returned CORRECT and 0 otherwise, and let c_i be the consistency fraction of answer i. The score of the answer is

$$S_i = (\Sigma_j v_{ij} + c_i) / (m + 1)$$

Each responding judge and the consistency fraction therefore carry an equal share of the score. With three judges, each of the four parts is worth 25 points, and because c_i takes three values, scores are multiples of 12.5 (0, 12.5, 25, … 100). If a judge fails, the shares are renormalised so that the maximum stays 100. The answer with the highest score is shown, and ties go to the first answer. The reported trust score is that score times 100.

Equal weights are a simplicity choice. We did not fit the weights to data, because no labelled development set was reserved for that purpose, and Section 4.5 shows that the equal-weight sum is not clearly better than a judge-only signal. The score also treats a judge vote and the consistency fraction as comparable, an assumption that has no direct support in the literature.

Table 3.4 shows how the rule works on illustrative cases. The numbers are arithmetic from the formula and are not results. A score of exactly 75 is the lowest that reaches the Medium label, and a single failed judge changes the grid of possible scores.

Judges responding	Judges correct	Consistency fraction	Tie-breaker	Trust score	Label
3	3	1.0	agrees	100	High
3	3	0.5	agrees	87.5	High
3	2	1.0	agrees	75	Medium
3	2	0.5	agrees	62.5	Low
3	3	1.0	disagrees	90	High
3	2	1.0	disagrees	65	Low
3	1	0	agrees	25	Low
2 (one failed)	1	1.0	agrees	66.7	Low
2 (one failed)	2	0.5	agrees	83.3	High

Table 3.4 — Worked examples of the scoring rule (illustrative arithmetic, not data).

3.6 Tie-Breaker

3.6.1 What the tie-breaker does

A further model, NVIDIA Nemotron 3.5 Lightning 30B, receives the question and only the winning answer and returns one verdict with the same prompt as the judges but without the search context. If its verdict differs from the majority of the three judges for that answer, the score is reduced by 10 points (not below 0), and the reason shown to the user adds a caution that an independent check disagreed. If the tie-breaker fails, the score is not changed. The tie-breaker makes one call per question, not three, because it checks only the winner.

3.6.2 Why a tie-breaker

We added it for three reasons. Its effect could be tested only indirectly (Section 4.6).

- **A further developer for the cases that matter.** Judges from three developers can still share an error (Kim et al., 2025). A further model from another developer adds one more opinion at the point where a wrong answer would otherwise be shown with high confidence. It cannot remove shared mistakes, but it gives the panel a chance to be contradicted.
- **Cross-checking by several models has a published benefit for factual accuracy.** Du et al. (2024) report that several model instances that propose and debate answers reduce factual errors and hallucinations. Their method mostly used copies of one model and a debate procedure, so we treat it as motivation for a second check, not as evidence for this particular design.
- **Low cost.** One call on the winner adds a small fraction of the roughly sixteen calls that a question needs.

The size of the penalty, 10 points, is a design choice that we did not tune. The evaluation file does not store the tie-breaker's verdict, but the penalty leaves a trace in the score. An unpenalised score is a multiple of 12.5, or a multiple of 16.7 when a judge failed, so a score that lies on neither grid must have been reduced by 10 points. Section 4.6 uses this to estimate how often the tie-breaker disagreed and what it did to the ranking. This is an inference from the scoring rule and not a logged event, and it does not replace a run with the tie-breaker switched off (Section 7).

3.7 Confidence Labels

A score of 80 or more is labelled High confidence, a score from 75 up to but not including 80 is labelled Medium confidence, and a lower score is labelled Low confidence. These cutoffs have not changed since the first commit of the repository (16 August 2026), which is before the seed-7 questions were drawn, and none was fitted to the seed-7 results. The repository also holds a script that explores thresholds on an earlier 200-question run (seed 42), so the cutoffs may have been informed by that run. The seed-7 sample excludes every one of its questions, so the labels are tested on questions that did not shape them, although we cannot rule out that the cutoffs were set with similar models and benchmarks in mind. Because the scores are multiples of 12.5 when all judges respond, the Medium band is in practice narrow. A score of 75 is Medium, 83.3 and above is High, and a penalised score such as 77.5 can also be Medium.

Decision	Main supporting evidence	Tested here?
Sample several answers and compare	Wang et al. (2023); Manakul et al. (2023)	Not isolated
LLM agreement check, not NLI	Manakul et al. (2023), Table 2; Kuhn et al. (2023) for the NLI alternative	No
Judges from three developers	Kim et al. (2025); Panickssery et al. (2024); Zheng et al. (2023)	Not isolated
Web search for the judges	Shuster et al. (2021); Li et al. (2023); Lewis et al. (2020)	No
Tie-breaker on the winner	Du et al. (2024); Kim et al. (2025)	Partly (Section 4.6, inferred)
Grader from a separate developer	Panickssery et al. (2024); Kamalloo et al. (2023)	No (not checked against people)
Equal weights, 10-point penalty	None (design choices)	Partly (Sections 4.5 and 4.6)

Table 3.2 — Design decisions, the evidence behind them and whether this paper tests them.

3.8 Evaluation Data

The test set has 200 questions drawn from three benchmarks: 66 from TruthfulQA (Lin et al., 2022), 66 from the question-answering part of HaluEval (Li et al., 2023) and 68 from the validation split of TriviaQA (Joshi et al., 2017). The set was sampled with random seed 7 and with every question of the earlier seed-42 run removed, so the two samples do not overlap. Each benchmark supplies a reference answer (the best answer for TruthfulQA, the right answer for HaluEval and the answer value for TriviaQA). The 200 questions were processed in four batches of 50, and the batches were merged into one file. Rows that failed during a batch, for example because of a rate limit, were re-run with the script's backfill option, and the merged file holds a score for all 200 questions. We use a single run and a single seed.

Benchmark	Questions	Reference	Role in the study
TruthfulQA	66	Best answer	Common misconceptions
HaluEval (QA)	66	Right answer	Hallucination detection
TriviaQA (validation)	68	Answer value	Factual recall
Total	**200**		

Table 3.3 — Composition of the 200-question evaluation set (seed 7).

3.9 Ground Truth, Metrics and Statistics

3.9.1 The grader and why it is a separate model

Whether an answer is correct is decided by Claude Haiku 5.5, which receives the question, the reference answer and the model's answer and is asked whether the answer conveys the same core truth as the reference, ignoring extra explanation, tone and length. It replies YES or NO at temperature 0, and the answer counts as correct only when the reply starts with YES. We use a language-model grader because exact matching against a gold answer underestimates generative models (Kamalloo et al., 2023), and because the answers here are free text of variable length.

The grader comes from Anthropic, a developer that plays no other role. Using a different developer for grading avoids the situation in which a model that took part in the scoring also decides what is correct, and so avoids the self-preference reported by Panickssery et al. (2024) and the self-enhancement bias described by Zheng et al. (2023). The code supports majority voting across several graders, but one grader was used. Kamalloo et al. (2023) report that automated evaluators cannot fully replace human evaluation, and we have not measured how often this grader agrees with people. Cohen's (1960) kappa on a human-labelled subsample is the planned check (Section 7). Until it is done, every accuracy and AUROC in this paper is conditional on the grader.

3.9.2 Metrics

- **Accuracy** is the share of questions whose chosen answer the grader marks correct.
- **Score gap** is the mean trust score of correct answers minus the mean trust score of incorrect answers.
- **AUROC** is the probability that a randomly chosen correct answer has a higher trust score than a randomly chosen incorrect one, with ties counted as half (Fawcett, 2006). It is computed from ranks. A value of 0.5 means no separation, and unlike the gap it does not depend on the scale of the score.
- **Brier score** (Brier, 1950) is the mean squared difference between the score divided by 100 and the outcome (1 for correct, 0 for incorrect). The reference value is the Brier score of always predicting the observed accuracy.
- **Expected calibration error** (Guo et al., 2017) is computed with five equal-width bins of the score.
- **Flagging analysis** treats a score below a threshold as a prediction that the answer is wrong, and reports the share of flagged answers that are wrong (precision), the share of wrong answers that were flagged (recall) and the accuracy of the answers that are not flagged.

Intervals are 95% percentile bootstrap intervals from 5,000 resamples of the 200 questions with a fixed seed (Efron and Tibshirani, 1993). The comparison of two scores on the same questions uses the same resamples for both, so the interval for the difference accounts for the

pairing. With only 200 questions the intervals are wide, and differences smaller than their width should not be read as findings.

3.10 Implementation, Cost and Deployment

The pipeline is written in Python with a one-node LangGraph graph and calls every provider through one function that speaks the OpenAI chat interface, with a separate client for Groq. A question needs three writer calls, two or three agreement calls, nine judge calls, one tie-breaker call and three web searches, about sixteen model calls and three searches in total, and one more call when the grader is used during evaluation. The calls to the judges are the largest part of the cost, which is the reason for keeping the number of samples at three.

The same code serves a public web page, https://trust.145-241-213-220.sslip.io, built with FastAPI. It runs on an Oracle Cloud ARM virtual machine (4 cores and 24 GB of memory, Ubuntu 22.04) behind a Caddy proxy that provides HTTPS. The service runs under systemd with a 4 GB memory limit so that it cannot starve other services on the same machine. To protect the model quotas, each visitor is limited to six questions per minute and at most eight verification runs may be in progress at once, with a short wait before a busy response is returned. A health route reports that the service is running. The source code is in the public repository listed under Declarations.

4. Results

4.1 Overview

All 200 questions produced an answer and a score. The writer's chosen answer was correct for 122 questions, an accuracy of 61.0% (95% interval 54.5–67.5). The mean trust score was 93.1 for correct answers and 74.9 for incorrect answers, a gap of 18.1 points (interval 11.1–25.5). The AUROC was 0.674 (interval 0.604–0.740). The interval excludes 0.5, so the score separates correct from incorrect answers better than chance, but the separation is moderate. Table 4.1 gives these and the calibration figures, which are discussed in Section 4.4.

Measure	Value
Questions	200
Accuracy of the chosen answer	61.0% [54.5, 67.5]
Mean trust score, correct answers	93.1
Mean trust score, incorrect answers	74.9
Score gap (correct minus incorrect)	18.1 [11.1, 25.5]
AUROC	0.674 [0.604, 0.740]
Answers with trust score 100	132 (37 of them wrong, 28.0%)
Answers with trust score 0	5 (all five wrong)
Brier score (reference: 0.238)	0.272
Expected calibration error (5 bins)	0.250

Table 4.1 — Main results on the 200 fresh questions (95% bootstrap intervals in brackets).

Figure 4.1 shows where the answers fall. Most answers, 132 of 200, received the maximum score, and 95 of those were correct. The score is discrete, because three judges and a three-valued consistency fraction produce multiples of 12.5, so the figure groups the values into five bands. The share of wrong answers is 73% in the 0–49 band (11 of 15) and 74% in the 50–74 band (17 of 23), 44% at exactly 75 (8 of 18), 42% in the 76–99 band (5 of 12) and 28% at 100 (37 of 132).

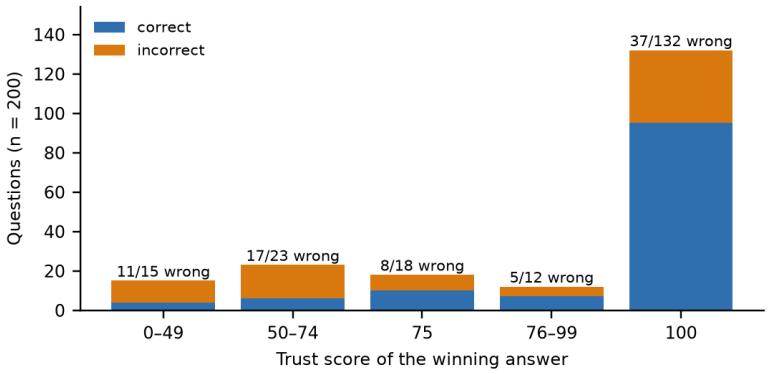


Figure 4.1 — Trust score of the chosen answer, grouped into five bands, split by whether the grader marked it correct. Labels give the number of wrong answers in each band.

4.2 Results by Benchmark

The separation differs by benchmark (Table 4.2 and Figure 4.2). TriviaQA gave the highest AUROC (0.700), with a gap of 24.9 points, and the writer was correct on 70.6% of its 68 questions. HaluEval gave an AUROC of 0.686 and a gap of 17.3, but the writer was correct on only 43.9% of its 66 questions, so this benchmark was the hardest for the writer. TruthfulQA gave the weakest separation, an AUROC of 0.568 and a gap of 9.3 points, even though the writer was correct on 68.2% of its questions. The interval for TruthfulQA (0.450–0.682) includes 0.5, so on this sample we cannot say that the score separates correct from incorrect answers on TruthfulQA at all. The intervals for all three benchmarks overlap, so the ordering of the benchmarks is suggestive and not established.

Benchmark	n	Accuracy	Mean trust, correct	Mean trust, wrong	Gap	AUROC [interval]
TriviaQA	68	70.6%	92.5	67.5	24.9	0.700 [0.564, 0.823]
HaluEval	66	43.9%	89.6	72.3	17.3	0.686 [0.564, 0.800]
TruthfulQA	66	68.2%	95.9	86.7	9.3	0.568 [0.450, 0.682]
All	**200**	**61.0%**	**93.1**	**74.9**	**18.1**	**0.674 [0.604, 0.740]**

Table 4.2 — Results by benchmark (AUROC with 95% bootstrap interval).

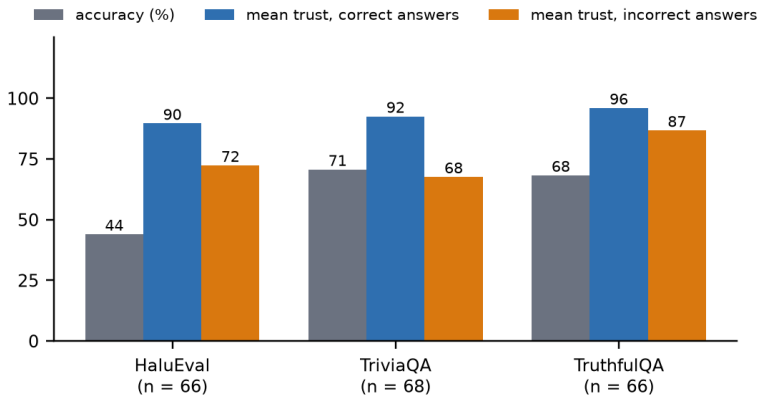


Figure 4.2 — Accuracy and mean trust score by benchmark. The gap between the blue and orange bars is the score gap.

The pattern for TruthfulQA is the one expected from the design of the benchmark. Lin et al. (2022) built the questions so that models imitate human falsehoods, and they found that larger models were generally less truthful. When a false belief is common in the training text, every model in the panel may hold it, which is the situation in which agreement and cross-model verdicts add little (Kim et al., 2025). Our data are consistent with this explanation, but they do not test it, because we did not analyse which questions the judges got wrong together.

4.3 Confidence Labels and Flagging

The labels were ordered as intended (Table 4.3 and Figure 4.3a). Of the 142 answers labelled High confidence, 71.1% were correct. Of the 20 labelled Medium, 55.0% were correct, and of the 38 labelled Low, 26.3% were correct. The High label is therefore not a guarantee, because more than one in four answers with that label were wrong. The Medium band holds only 20 answers, so its 55.0% rests on a small number and has a wide interval.

Label	Score range	Answers	Share of all answers	Correct
High confidence	80 or more	142	71.0%	71.1%
Medium confidence	75 to below 80	20	10.0%	55.0%
Low confidence	below 75	38	19.0%	26.3%

Table 4.3 — Accuracy by confidence label.

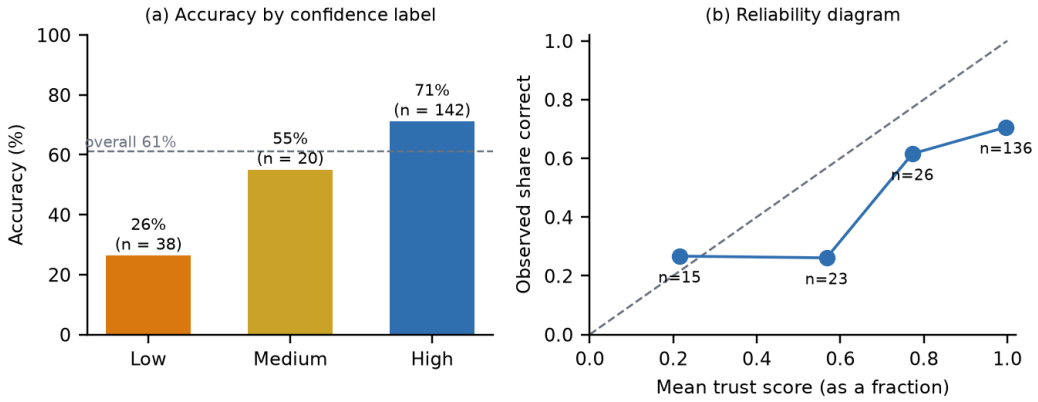


Figure 4.3 — (a) Accuracy for each confidence label. (b) Reliability diagram with four bins of the trust score. The dashed line is perfect calibration.

Table 4.4 treats a score below a threshold as a prediction that the answer is wrong. With the strictest rule, flagging everything below 100, 68 answers are flagged and 60.3% of them are wrong. Flagging at the Low cutoff (below 75) flags 38 answers, and 73.7% of those are wrong, but only 35.9% of all wrong answers are caught. The accuracy of the answers that are not flagged is between 64% and 72% for every threshold, against 61.0% overall, so filtering by score improves the accuracy of what remains by about eleven points at most.

Flag if score below	Flagged	Wrong among flagged	Wrong answers caught	Accuracy of the rest
100	68	60.3%	52.6%	72.0% (132 kept)
80	58	63.8%	47.4%	71.1% (142 kept)
75	38	73.7%	35.9%	69.1% (162 kept)
66.7	34	79.4%	34.6%	69.3% (166 kept)
50	15	73.3%	14.1%	63.8% (185 kept)

Table 4.4 — Flagging answers whose trust score is below a threshold as likely wrong.

4.4 Calibration

The score should not be read as a probability. The Brier score of the trust score divided by 100 was 0.272, which is worse than 0.238, the Brier score of a constant forecast equal to the observed accuracy of 61%. The expected calibration error with five equal-width bins was 0.250. Figure 4.3b shows why. The 136 answers with a mean score of 0.997 were correct 70.6% of the time, and the 23 answers with a mean score of 0.57 were correct only 26.1% of the time, no better than the 15 answers with a mean score of 0.22 (26.7%). The score orders the answers but its numerical value is too high, which is the kind of miscalibration that Guo et al. (2017) describe for modern models and that Xiong et al. (2024) report for verbalised confidence.

4.5 Which Signal Carries the Information?

The saved results do not contain the per-judge votes, the consistency fraction or the tie-breaker verdict, so the components of the score cannot be separated exactly. They do contain, for each question, how many of the three generated answers were flagged as hallucinated by at least one judge. The share of answers that no judge flagged is a signal that uses judge verdicts only, over all three answers, with no consistency fraction and no tie-breaker. It is a proxy, not a clean ablation, because it uses the verdicts on all three answers and not only on the winner.

This judge-flag proxy ranked answers at least as well as the full trust score (Table 4.5). Its AUROC was 0.721 (interval 0.655–0.782) against 0.674 for the trust score, a difference of +0.046 whose paired interval (−0.005 to +0.098) just includes zero. The data therefore do not show that adding the agreement check and the tie-breaker improved the ranking, and a reader should not conclude from this table that they hurt it either. Accuracy fell steadily as judges flagged more of the three answers (Table 4.6). When no judge flagged any answer (138 questions), 75.4% of the chosen answers were correct. When all three answers were flagged (22 questions), only 13.6% were.

Signal	**AUROC [95% interval]**
Trust score (agreement, judges, tie-breaker)	0.674 [0.604, 0.740]
Judge-flag proxy (answers no judge flagged)	0.721 [0.655, 0.782]
Difference, proxy minus trust score (paired)	+0.046 [−0.005, +0.098]

Table 4.5 — Ranking quality of the trust score and of a judge-only proxy on the same 200 questions.

Answers no judge flagged	**Questions**	**Chosen answer correct**
0 of 3	22	13.6%
1 of 3	15	33.3%
2 of 3	25	40.0%
3 of 3	138	75.4%

Table 4.6 — Accuracy by the number of the three answers that no judge flagged.

Two simple flagging rules based on the same field give a comparison with Table 4.4. Flagging a question when any judge flagged any of the three answers flags 62 questions, 71.0% of which are wrong, and catches 56.4% of the wrong answers. Flagging when more than half of the answers were flagged flags 37 questions, 78.4% of which are wrong, and catches 37.2% of the wrong answers. The first rule flags about as many questions as the strictest score rule in Table 4.4 (62 against 68) and is more precise (71.0% against 60.3%) with a similar recall (56.4% against 52.6%).

4.6 What the Scores Reveal About the Tie-Breaker and Failed Judges

Section 3.5 gives the grids on which scores can fall. With three judges an unpenalised score is a multiple of 12.5. With one failed judge it is a multiple of 16.7. A score on neither grid must have had the 10-point tie-breaker penalty applied, and a score on the second grid only, such as 33.3, 66.7 or 83.3, shows that a judge failed and the shares were renormalised. Reading the saved scores this way gives the counts in Table 4.7. It relies on the scoring rule being as described and on no other step changing the score, and it is an inference, not a logged event.

The tie-breaker disagreed with the judge majority on 17 of the 200 answers (8.5%). Only 6 of those 17 answers (35.3%) were correct, against 116 of the other 183 (63.4%), a difference of 28.1 points whose interval (2.3 to 51.6) excludes zero. Adding the 10 points back to those 17 scores lowers the AUROC from 0.674 to 0.661, so the penalty raised the AUROC by 0.013 (paired interval −0.003 to +0.033). The tie-breaker therefore flagged a group of answers that were mostly wrong, but its overall effect on the ranking is small and cannot be told apart from zero with 200 questions. Eleven of the 17 penalised answers received a Low label, four High and two Medium. At least 12 answers (6.0%) have scores that arise only when a judge failed. The true number is larger, because a failure that leaves the score on the main grid, such as 50 or 100, cannot be seen.

Quantity	Value
Answers with the 10-point penalty	17 of 200 (8.5%)
Correct among penalised answers	6 of 17 (35.3%)
Correct among other answers	116 of 183 (63.4%)
Difference in accuracy (other minus penalised)	+28.1 points [2.3, 51.6]
AUROC with the penalty / with it added back	0.674 / 0.661
Effect of the penalty on AUROC (paired)	+0.013 [−0.003, +0.033]
Score gap with the penalty / with it added back	18.1 / 17.2
Answers whose score shows a failed judge (lower bound)	12 of 200 (6.0%)

Table 4.7 — Tie-breaker penalties and failed judges, inferred from the saved scores.

4.7 Summary of Findings

- **RQ1.** The score separates correct from incorrect answers moderately well (AUROC 0.674, interval 0.604–0.740; gap 18.1 points, interval 11.1–25.5).
- **RQ2.** The separation is strongest on TriviaQA (0.700) and HaluEval (0.686) and not distinguishable from chance on TruthfulQA (0.568, interval 0.450–0.682), although the intervals overlap.
- **RQ3.** The labels are ordered (71.1%, 55.0% and 26.3% correct for High, Medium and Low), but the score is not a calibrated probability, and 28.0% of answers with the maximum score were wrong.
- **RQ4.** A judge-only proxy ranked at least as well as the full score (0.721). The tie-breaker disagreed on 8.5% of answers, mostly wrong ones, but added only +0.013 to the AUROC

(interval −0.003 to +0.033). The effects of the agreement check and of web search were not isolated and remain open.

5. Discussion

5.1 What the Results Support

The main supported statement is a modest one. On 200 fresh questions the trust score ranked correct answers above incorrect ones more often than chance, with an AUROC near 0.67, and the three labels sorted the answers into groups with clearly different accuracy. That is useful for triage. A user who sees a Low label has good reason to check the answer, because about three in four such answers were wrong. The results do not support stronger statements. The score does not make the answer more accurate, since the layer only chooses among three samples from the writer, and we did not measure whether that choice beats a random sample. The score is not a probability. And a High label leaves more than a quarter of the answers wrong.

5.2 Why High Scores Can Still Be Wrong

Of the 132 answers with the maximum score, 37 were wrong (15 from TruthfulQA, 13 from HaluEval and 9 from TriviaQA). Three explanations are consistent with the design and with the literature. The first is shared belief. Kim et al. (2025) found that models often make the same wrong choice, and that the correlation does not disappear across providers, so three judges and a Gemini checker can all accept a false claim that the writer also believes. Agreement between samples cannot expose it, because a model can be consistently wrong, as Farquhar et al. (2024) note when they restrict their claim to confabulations. The second is abstention. The writer was told not to invent details, and when it replies that it cannot find information about a person, the judges tend to count the reply as not hallucinated, while the grader compares it with the reference and marks it incorrect. The third is the reference itself, which for some questions is ambiguous or follows a convention that the answer does not.

To illustrate these three explanations, we read nine of the 37 wrong top-scoring answers chosen at random (random seed 3). This is an informal reading and not a measured result. Three were abstentions, such as a reply that no information could be found about a named person. Three involved an ambiguous or conventional reference, for example a question about the Nobel Peace Prize in 1920, where the answer names one laureate and the reference names a different person. Three were plausible but wrong or hedged answers, such as one that repeats a popular but disputed meaning of a phrase. These examples suggest that part of the apparent error rate comes from the single LLM grader and the references, not only from the pipeline, which is why Section 3.9 treats every accuracy as conditional on the grader.

5.3 The Judge-Only Result

The judge-flag proxy matched or exceeded the full score in this sample, and the paired difference was not statistically distinguishable from zero. This does not show that agreement or the tie-breaker are useless. The proxy sees the verdicts on all three answers, and the trust score sees the verdicts on each answer separately and then picks the winner, so the two are not the same quantity. The result is a reason for follow-up, not for a design change. One plausible explanation for a weak agreement signal is that three samples give the consistency fraction only three values, while Manakul et al. (2023) report results with larger numbers of samples (their main results use 20). Whether more samples would help here is untested. The tie-breaker, in contrast, flagged a group of answers that were mostly wrong (Section 4.6), although its effect on the ranking was small.

5.4 Use in Practice

The results suggest three rules for presenting the score. First, show the label and the reasons, such as how many judges confirmed the answer and whether the answers agreed, and avoid presenting the number as a probability. Second, treat the Low label as a request to verify and the High label as the absence of a warning, not as a guarantee. Third, do not rely on the score for questions about common misconceptions, where our evidence of separation is weakest. The public page follows the first rule by showing the judges' verdicts and any tie-breaker caution next to the score.

5.5 Relation to Earlier Work

The AUROC values of this study cannot be compared directly with published ones. Xiong et al. (2024) report AUROC values for failure prediction on other models and datasets (0.522 for black-box and 0.605 for white-box methods in one comparison), Kuhn et al. (2023) and Farquhar et al. (2024) use semantic entropy with open models, and Manakul et al. (2023) evaluate passage-level detection with a different metric. What the present results add is a measurement of a layer that combines several of these ideas, on questions from three benchmarks, with intervals.

6. Limitations and Threats to Validity

Every result above has to be read with the following limits.

- **The grader is a model, not a person.** Correctness is decided by Claude Haiku 5.5 against a reference answer. We did not measure its agreement with human labels, which Kamalloo et al. (2023) identify as the safeguard that automated evaluation still needs. Section 5.2 shows cases where the reference or the grader, and not the pipeline, may explain a wrong label. Every accuracy and AUROC is conditional on the grader.
- **Small sample and a single run.** There are 200 questions, one seed and one run, so the intervals are wide and many differences are within noise. The per-benchmark

intervals overlap. Model outputs at temperature 0.9 are not reproducible bit for bit, and hosted models change over time, so a repeat run may give different numbers.

- **Judges are not independent.** The three judges come from three developers, but Kim et al. (2025) show that accurate models from different providers still make correlated errors. The 28.0% of wrong answers at the maximum score and the weak TruthfulQA result are consistent with this.
- **Agreement is not truth.** The consistency fraction measures whether the writer repeats itself. A consistently wrong writer scores full consistency. With three samples it takes only three values.
- **The agreement prompt leans towards agreement.** The prompt tells the checker that extra detail is not disagreement. That is intended, but it can raise the consistency score for answers that differ in substance.
- **The components were not isolated.** There is no run with web search, the tie-breaker or the agreement check switched off, and the saved results do not store per-judge votes, the consistency fraction or the tie-breaker verdict. The effect of the tie-breaker is inferred from the scores (Section 4.6). The Gemini checker was not compared with an NLI checker on our data. The effect of web grounding on the judges was not measured. These are design choices supported by indirect evidence (Table 3.2).
- **Equal weights and the penalty are untuned.** The equal weights and the 10-point penalty were not fitted, and the judge-only proxy suggests the sum may not be optimal.
- **Cutoffs.** The 80 and 75 cutoffs have not changed since 16 August 2026 and none was fitted to the seed-7 results, but a script that explores thresholds on an earlier run exists, so we cannot call the label evaluation fully out-of-sample.
- **Failures are not logged per question.** The evaluation file does not record rate-limit retries or dropped judges for each question. When a judge is dropped the remaining shares are renormalised, so a failure can change a score without leaving a trace. From the score arithmetic we can see that at least 12 answers (6.0%) had a judge fail, and the true number is larger.
- **Scope of the benchmarks.** The three benchmarks have short factual questions with reference answers. The results say nothing about long answers, other languages, or domain questions in medicine or law, where the motivation for the work is strongest.
- **Supporting studies are not on our task.** Several cited studies were run on summarisation, reasoning, chat preferences or dialogue (Panickssery et al., 2024; Huang et al., 2024; Zheng et al., 2023; Shuster et al., 2021). They motivate the design and do not prove that each effect occurs for short factual verdicts.

7. Conclusion and Future Work

7.1 Summary of Contributions

This paper described LLM Trust Score, a black-box layer that scores a language-model answer by combining agreement between three sampled answers, the verdicts of three judges from three developers grounded by web search, and a check of the winning answer by a further model. On 200 fresh questions the score separated correct from incorrect answers moderately well (AUROC 0.674, interval 0.604–0.740), and its High, Medium and Low labels were ordered (71.1%, 55.0% and 26.3% correct). It was weakest on TruthfulQA, it was not calibrated, and a quarter of the answers with the maximum score were wrong. We explained why each component was chosen, including why a Gemini agreement checker was preferred to an NLI model on the evidence of Manakul et al. (2023), and we stated which components were not evaluated on their own.

7.2 Recommendations for Future Work

The following experiments would turn the open design questions into measured results, in priority order.

- **Check the grader against people.** Label about 100 answers by hand and report agreement and Cohen's (1960) kappa for the grader, with a second grader from another developer for comparison.
- **Run ablations on the same questions.** Switch off, one at a time, the agreement check, web search and the tie-breaker, and use a single judge, and store per-judge votes and the consistency fraction so that the components can be analysed.
- **Compare checkers.** Label 30 answer sets by hand and compare the Gemini checker with the DeBERTa NLI checker of Kuhn et al. (2023) on accuracy, repeatability and cost.
- **Repeat with more questions and seeds.** Use several seeds and a larger sample so that the benchmark differences can be tested, and run the full TruthfulQA and HaluEval question-answering sets.
- **Vary the number of samples.** Test five and ten samples against three, following Manakul et al. (2023).
- **Fit the cutoffs and the weights on one half and test on the other,** or use cross-validation, and test a calibration map of the score to a probability on held-out data.
- **Test on harder, domain-specific questions** where the cost of a wrong answer is high.

Declarations

Data and code availability. The code, the evaluation scripts and the results file for the seed-7 run (`evaluation/results_seed7_final.csv`) are in the project repository, github.com/kkhushbuj/Hallucination-Detection-Reliable-Layer. The benchmark data are the public TruthfulQA, HaluEval and TriviaQA datasets. The model identifiers are those recorded in `config.py` at the time of the run (29 October 2026). A public web page serves the same pipeline at https://trust.145-241-213-220.sslip.io.

Ethics. The study used public benchmark data and model outputs and involved no human participants and no personal data.

Use of AI tools. Language models are the object of this study and are used inside the pipeline as described. An AI assistant also helped to draft the text of this paper and to compute the statistics from the saved results file. The author is responsible for the content.

Funding and competing interests. No external funding was received. The author declares no competing interests.

Appendix A. Prompts Used in the Pipeline

The prompts are reproduced from the source code, with typographic dashes and arrows simplified. Text in braces is filled in at run time. Every call also carries the system instruction in the first block.

```
SYSTEM INSTRUCTION (sent with every model call)
If the user asks about something that does not exist (a fake study, fake person, fake event,
fake citation, etc.), clearly state that it does not exist and STOP there. Do not invent
hypothetical details, fake findings, fake quotes, or fictional elaboration, even if you
label it as imaginative or hypothetical.
```

```
WRITER (three calls, temperature 0.9)
{question}
```

```
AGREEMENT CHECK (Gemini 3.5 Flash-Lite, temperature 0)
Answer A: {answer_a}

Answer B: {answer_b}

Do these two answers reach the SAME final conclusion? IMPORTANT: one answer may include
extra explanation, steps, or detail that the other doesn't have; this does NOT count as
disagreement. Only judge based on the final conclusion/result itself.

Example: 'The answer is 12.' and 'To solve this, multiply 80 by 0.15, which equals 12.' ->
these AGREE, both conclude 12.

Respond with ONLY the single word YES or the single word NO. Do not include any explanation
or other text.
```

```
JUDGE (three models, temperature 0)
Question: {question}

Proposed answer: {answer}

Web search context: {summary and top three results}      (omitted when the search fails)

Based on your own knowledge and the web search context above, is this answer factually
correct and not hallucinated? Respond with exactly one word: CORRECT or HALLUCINATED.
```

```
TIE-BREAKER (Nemotron 3.5 Lightning, temperature 0, winning answer only, no search context)
Question: {question}

Proposed answer: {winning answer}

Based on your own knowledge, is this answer factually correct and not hallucinated? Respond
with exactly one word: CORRECT or HALLUCINATED.
```

```
GRADER (Claude Haiku 5.5, temperature 0, evaluation only)
Question: {question}

Expected correct answer: {reference}

Model's actual answer: {answer}

Does the model's answer convey the same core truth/conclusion as the expected correct
answer, even if worded very differently or with more detail? Ignore extra explanation, tone,
or length. Focus only on whether the core factual conclusion matches.

Respond with exactly one word: YES or NO.
```

Appendix B. Distribution of Trust Scores

Table B.1 lists every score that occurred in the seed-7 run. Scores such as 15, 40, 65, 77.5 and 90 lie off the grid of unpenalised scores and are the ones attributed to the tie-breaker penalty in Section 4.6. Scores such as 33.3, 66.7 and 83.3 arise when a judge failed.

Trust score	Answers	Correct	Wrong	Share correct
0	5	0	5	0.0%
15	1	0	1	0.0%
25	3	1	2	33.3%
33.3	1	0	1	0.0%
40	5	3	2	60.0%
50	12	2	10	16.7%
56.7	2	0	2	0.0%
62.5	3	1	2	33.3%
65	2	0	2	0.0%
66.7	3	2	1	66.7%
73.3	1	1	0	100.0%
75	18	10	8	55.6%
77.5	2	1	1	50.0%
83.3	5	4	1	80.0%
87.5	1	1	0	100.0%
90	4	1	3	25.0%
100	132	95	37	72.0%
Total	**200**	**122**	**78**	**61.0%**

Table B.1 — Number of answers and correct answers at each trust score (seed-7 run).

Appendix C. Reproducing the Results

The evaluation uses the scripts in the repository. Each command needs the provider keys in a local environment file. The run for this paper used the seed-7 sample.

```
python evaluation/evaluate.py --seed 7 --batch 1     # repeat for --batch 2, 3 and 4
python evaluation/evaluate.py --seed 7 --backfill    # re-run failed rows, then merge
python evaluation/evaluate.py --seed 7 --merge       # write results_seed7_final.csv
```

The statistics in Section 4 are computed from `results_seed7_final.csv`, which has one row per question with the columns question, source, model_answer, best_answer, trust_score, label, flagged_count, majority_flagged and correct. The bootstrap uses 5,000 resamples with random seed 7. Because the models are hosted and the writer samples at temperature 0.9, a new run will not reproduce the same answers exactly.

Appendix D. Evidence Behind the Main Statements

Table D.1 lists the main statements about the world that the paper makes, the source, and what the source actually covers. Statements about our own system rest on our own data (Section 4) or are labelled as design choices.

Statement	Source	Scope of the source
Calibrated models must hallucinate on facts seen once	Kalai and Vempala (2024)	Theory; hallucination not inevitable overall
Confabulation is flagged by variation over meaning	Farquhar et al. (2024)	One error type
Sampling several answers and taking the most frequent helps	Wang et al. (2023)	Reasoning tasks, four models
Facts recur across samples, invented ones diverge	Manakul et al. (2023)	GPT-3 passages about people
Prompted LLM beat NLI for consistency scoring	Manakul et al. (2023), Table 2	Passage and sentence level, not short answers
Meaning classes by bidirectional entailment	Kuhn et al. (2023)	DeBERTa-large MNLI, QA datasets
Black-box confidence close to white-box	Xiong et al. (2024)	Five dataset types, five models
Stated confidence is better calibrated than token probabilities	Tian et al. (2023)	RLHF models on three benchmarks
LLM judges agree with people over 80%	Zheng et al. (2023)	Chat preferences
Judges favour their own writing	Panickssery et al. (2024); Zheng et al. (2023)	Summarisation; chat
Models cannot self-correct reasoning without feedback	Huang et al. (2024)	Reasoning
Errors correlate across models and providers	Kim et al. (2025)	350+ models; leaderboards
Models imitate human falsehoods; larger ones less truthful	Lin et al. (2022)	817 questions, GPT-3 era models
Retrieval cuts hallucination	Shuster et al. (2021)	Dialogue, Wikipedia

Statement	Source	Scope of the source
External knowledge helps detection	Li et al. (2023)	HaluEval
Lexical matching underestimates generative QA	Kamalloo et al. (2023)	Open-domain QA
Several instances debating reduce factual errors	Du et al. (2024)	Mostly one model family

Table D.1 — Statements, their sources and the scope of each source.

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